THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Pressure, Winds, Storms and Cyclones** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(c)** 3822 Pa	Area = 3.14 x 0.01 = 0.0314 square m; 120/0.0314 = 3822 Pa.
3	**(b)** A is likely to be stronger	Sea supplies energy.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(c)** 4000 Pa	Inversely proportional.
6	**(a)** wind and waves are strongest near the eye wall	Danger.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(d)** 60 N upward	Balanced forces.
9	**(b)** reduce the force of the surge	Wave barrier.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Smaller area gives higher pressure.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Area = 0.005 square metre (1); P = 50/0.005 = 10000 Pa (1).	1+1
12	They show cloud patterns over a large area (1); Help in predicting path and intensity (1).	1+1
13	Wind lowers pressure above the roof (1); Higher pressure inside pushes it up (1).	1+1
14	P = 450/0.0009 = 500000 Pa (1); Very high; can damage floors (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area = 0.01 square m; P = 50/0.01 = 5000 Pa (1); Stacked: weight 100 N (1); P = 10000 Pa (1).	1+1+1
16	Pressure gradient is steepest (1); Air rushes toward the centre (1); Rotation increases speed (1).	1+1+1
17	Helium is lighter than air (1); Surrounding air provides upward push (1); Air-filled balloon is not lighter (1).	1+1+1
	OR: Feet: 700/0.04 = 17500 Pa (1); Skis: 700/0.4 = 1750 Pa (1); Skis reduce pressure to one-tenth (1).	1+1+1
18	Falls on approach (1); Lowest at the eye (1); Rises after it passes (1).	1+1+1
	OR: Crops are damaged (1); Soil erosion (1); Early warnings and insurance (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Pressure at 3 m is three times that at 1 m (1); Ears feel more pressure at depth (1); Dam base is thick (1); Pressure acts in all directions (1); Independent of container shape (1).	1+1+1+1+1
	OR: Pushing in reduces volume, pressure rises (1); More collisions with walls (1); Pulling out lowers pressure (1); Outside air enters the pump (1); Pump compresses air into tyre (1).	1+1+1+1+1
20	Cow: area 0.04 square metre (1); Cow pressure = 4000/0.04 = 100000 Pa (1); Man = 800/0.04 = 20000 Pa (1); Cow's pressure is five times (1); Cow may sink in wet soil more easily (1).	1+1+1+1+1
	OR: Equator heated more (1); Hot air rises (1); Cool polar air moves toward equator (1); Rotation of the Earth deflects (1); Global wind patterns result (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Pressure, Winds, Storms and Cyclones** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 0.02 square metre	800 / 40000.
2	**(a)** less air lies above	Smaller air column.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Energy supply cut.
4	**(a)** 1000000 Pa	500/0.0005.
5	**(b)** storm surge	Low pressure and strong winds push water.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(b)** twice	Proportional to depth.
8	**(c)** anemometer	Measures wind speed.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(d)** outside pressure falls and the balloon expands	Balloon swells.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rotation of the Earth deflects winds (1); Spiralling inward gives rotation (1).	1+1
12	Pressure and oxygen are low (1); The body adapts slowly (1).	1+1
13	Hot air expands, becomes less dense (1); Cooler air flows in to replace it giving wind (1).	1+1
14	Outside pressure is lower (1); Air inside expands (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Air is pushed out leaving low pressure under the cup (1); Outside pressure holds it (1); Cracks let air in, so the seal fails (1).	1+1+1
16	Pressure change is large (1); Gases in the body expand (1); Body structure is damaged (1).	1+1+1
17	Snowshoes increase the area (1); Pressure on snow falls (1); Less sinking (1).	1+1+1
	OR: Pressure is proportional to depth (1); 9/3 = 3 times (1); Pressure = 3P (excluding air) (1).	1+1+1
18	Very low pressure inside (1); Fast rotating winds (1); Buildings explode and objects lift (1).	1+1+1
	OR: Atmospheric pressure pushes the card up (1); It exceeds the weight of the water (1); Seal prevents air entry (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Warm moist air rises (1); Low pressure draws air spiral (1); Eye and eye wall (1); Over land it loses moisture and weakens (1); Warm ocean supplies energy and vapour (1).	1+1+1+1+1
	OR: All involve low pressure and rising air (1); Thunderstorm: local, short (1); Cyclone: large, over sea, days (1); Tornado: narrow, minutes (1); Damage patterns differ (1).	1+1+1+1+1
20	Collapsing can or inverted glass with card (1); Pressure outside exceeds inside (1); Straw or syringe (1); Suction cup (1); Application: pressure cooker or vacuum cleaner (1).	1+1+1+1+1
	OR: Pressure increases with depth (1); Thick strong walls (1); Pressure acts in all directions (1); Leak-proof seals (1); Check systems before dive (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Pressure, Winds, Storms and Cyclones** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** burst outwards	Pressure imbalance.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(a)** 2000 N	200000 x 0.01.
4	**(a)** moist air rises and cools there	Orographic rain.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(d)** there is less air above the hill station	Air column is shorter.
7	**(d)** air pressure and oxygen are low	Thin air.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(b)** an approaching storm	Low pressure.
10	**(d)** increases	Particles collide with the walls more often.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The wind returns from the other side (1); People may leave shelters too early (1).	1+1
12	Difference = 55 kPa (1); F = 55000 x 0.1 = 5500 N (1).	1+1
13	Pressure is proportional to depth (1); It is four times (1).	1+1
14	Large pressure at the base (1); Curved thick wall resists the push (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Summer: land hot, low pressure; wind from sea to land (1); Brings moist air and rain (1); Winter: land cold; wind from land to sea (1).	1+1+1
16	Water pressure at depth is large (1); Thick walls withstand it (1); Crew safe (1).	1+1+1
17	Before: warm, humid, clouds build (1); During: lightning, rain, strong winds (1); After: cooler, clearer air (1).	1+1+1
	OR: Difference = 6 kPa (1); Air flows from high to low pressure (1); Strong inward winds (1).	1+1+1
18	4 N per square cm (1); 12 N / 0.0003 square m (1); = 40000 Pa (1).	1+1+1
	OR: Charge may travel through wires (1); Risk of electric shock (1); Surges can damage appliances (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Satellite tracking (1); Barometers and anemometers (1); Colour-coded alerts (1); Cyclone shelters and evacuation (1); Coastal embankments and mangrove belts (1).	1+1+1+1+1
	OR: Weight of air column (1); Straw (1); Suction cup (1); Collapsing can (1); Balloons and tyres (1).	1+1+1+1+1
20	Land heats quickly (1); Hot air rises, low pressure (1); Sea air flows inland (1); Night: land cools, sea warmer (1); Land breeze towards the sea (1).	1+1+1+1+1
	OR: Weather bulletins (1); Mock drills (1); Safe shelter identification (1); Emergency kit (1); Lightning safety rules (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.